

Curriculum Vitae

Lien Cartaya Bermudez

Ph.D. Candidate in Mathematics

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Research Interests

Representation theory, reflection groups, invariant theory, rational Cherednik algebras, non-commutative algebra, path algebras, combinatorics, and computational algebra.

Publications

- L. Cartaya and S. Griffeth. *Zero fibers of quaternionic quotient singularities*. *Revista Matematica Iberoamericana* 41 (2025), no. 3, 1461–1487. DOI: [10.4171/RMI/1528](https://doi.org/10.4171/RMI/1528).

Manuscripts in Preparation

- L. Cartaya. *Zero fiber rings for rank 2 quaternionic reflection groups*. In preparation.
- L. Cartaya, S. Griffeth, D. Guevara. *Wonderful models of subspace arrangements for division rings*. In preparation.

Education

2024–2028 (expected)	Ph.D. in Mathematics , Universidad de Talca, Chile. Advisor: Stephen Griffeth. Ph.D. Candidate since January 2025.
2022–2024	M.Sc. in Mathematics , Universidad de Talca, Chile. Thesis: <i>Non-commutative deformations of quotient singularities</i> (review of an article). Advisor: Stephen Griffeth.
2016–2020	Degree in Mathematics , Universidad de La Habana, Cuba. Thesis: <i>Study on high-dimensional algorithms for the reduction of STRESS in Multidimensional Scaling</i> . Advisor: Elina Miret Barroso.

Awards, Scholarships, and Selected Programs

2026	Selected young researcher, 13th Heidelberg Laureate Forum , Heidelberg, Germany.
2026	Operational expenses for research, ANID, Chile.
2024–2028	ANID Beca de Doctorado Nacional, Chile.

Appointments and Professional Experience

Sep 2026–	Mathematics Tutor, IDI Tutoring Academy . Online mathematics tutoring in English.
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2024–2026	Social Media Assistant (part-time), Institute of Mathematics, Universidad de Talca.
2023–2026	Thesis student, FONDECYT Regular Project 1231355, Chile.
2021	Data Scientist (part-time), Mandao, La Habana, Cuba.

Selected Conference Talks and Posters

2026

Sep	Oral: <i>What zero remember?</i> Next Gen Session, 13th Heidelberg Laureate Forum , Heidelberg, Germany.
Aug	Oral: <i>Zero fiber rings for rank 2 quaternionic reflection groups</i> . XXV Latin American Congress of Algebra (CLA) , Lima, Peru.
Jul	Oral: <i>A Coxeter-type bound for zero-fiber rings of quaternionic quotient singularities</i> . CIME School on Algebraic and geometric combinatorics: Coxeter group theory and representation theory , Cetraro, Italy.
Apr	Oral: <i>On bounds for the degree of the zero fiber of rank-2 quaternionic reflection groups</i> . XXXVIII Jornada de Matematica de la Zona Sur , Universidad del Bio-Bio, Concepcion, Chile.

2025

Aug	Poster: <i>The diagonal coinvariant ring for quaternionic reflection groups</i> . X Encuentro Nacional de Algebra – EMALCA Argentina, Cordoba, Argentina.
Apr	Oral: <i>About the diagonal coinvariant ring for quaternionic reflection groups</i> . XXXVII Jornada de Matematica de la Zona Sur, Universidad Catolica del Maule, Talca, Chile.

2024

Jul	Oral: <i>Zero fiber of quotient singularities</i> . XXIV Coloquio Latinoamericano de Algebra, Santiago de Chile, Chile.
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2021

Jul	Poster: <i>Application of multidimensional scaling to SARS-CoV-2 patients treated with Interferon</i> . I Festival Nacional de Matematica y Computacion, La Habana, Cuba.
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2020

Mar	Oral: <i>Study and implementation of algorithms for high-dimensional multidimensional scaling</i> . IX Encuentro de Matematica Numerica y Optimizacion, Cuba-Mexico, La Habana, Cuba.
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Seminar and Student Talks

2026

Jun	Oral: <i>Anillo de la fibra cero para grupos de reflexiones en rango 2</i> . Seminario de Estudiantes 2026, Universidad de La Frontera, Temuco, Chile.
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2025

Oct	Oral: <i>Walking along quivers towards finitely generated algebras</i> . Seminario de Estudiantes de la Universidad de Chile (Seminario Relajao), Santiago de Chile,
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	Chile; Seminario de Estudiantes de la Universidad de Valparaiso (Puerto de Matematicas), Valparaiso, Chile.
Apr	Oral: <i>Understanding reflection groups from their representations</i> . Seminario No-Degenerado, Instituto de Matematicas, Universidad de Talca, Chile.
2024	
May	Oral: <i>Quivers: a preliminary vision</i> . Seminario No-Degenerado, Instituto de Matematicas, Universidad de Talca, Chile.
Mar	Oral: <i>Zero fiber of quotient singularities</i> . 2do Encuentro Nacional de Estudiantes de Matematicas, Instituto de Matematicas, Universidad de Talca, Chile.
2023	
Aug	Oral: <i>McKay correspondence</i> . Seminario No-Degenerado, Instituto de Matematicas, Universidad de Talca, Chile.

Teaching Experience

Sep 2026–	Mathematics Tutor, IDI Tutoring Academy . Online mathematics tutoring in English.
Jun 2026	Instructor, online undergraduate course on <i>Groebner Bases</i> , Club de Matematicas Yachay Tech. Seven hours of classes and approximately 25 hours of academic preparation.
Mar 2026	Instructor, Preparatory Mathematics course for engineering degrees, Universidad de Talca. Duration: one week.
2025	Lecturer, Universidad de Talca. Precalculus and Calculus for Commercial Engineering.
2022–2025	Teaching Assistant, Universidad de Talca. Statistics and Probability, Calculus, Linear Algebra, ODEs.
2017–2022	Teaching Assistant, Universidad de La Habana. Calculus, Linear Algebra, Real Analysis.
Jan 2023	Teaching Assistant, Summer School on finite subgroups of SL_2 by S. Griffeth, Universidad de Talca.

Academic Service and Outreach

Jan 2026	Instructor of the short course <i>Invisible Patterns: Changing without changing</i> , Campamento Hannah, Instituto de Matematicas, Universidad de Talca, Chile. Mathematics outreach program for female high school students.
Jan 2026	Member of the Organizing Committee of the 2026 Summer School for Undergraduates in Mathematics, Instituto de Matematicas, Universidad de Talca, Chile.

Schools, Workshops, and Conferences Attended

Sep 2026	IA en la práctica matemática (IAPMat 2026) , Pontificia Universidad Católica de Chile, Santiago, Chile.
Jul 2026	Algebraic Combinatorics and Finite Groups IV , Cetraro, Italy.
Jun–Jul 2026	CIME School: Algebraic and geometric combinatorics: Coxeter group theory and representation theory , Cetraro, Italy.

Mar 2023

CIMPA School *Crossroads of Geometry, Representation Theory and Higher Structures*, Puerto Madryn, Argentina.

References

- Stephen Griffeth, Professor of Mathematics, Universidad de Talca, Chile. Email: sgriffeth@utalca.cl
- Maria Ofelia Ronco, Universidad de Talca, Chile. Email: mronco@utalca.cl

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